SwasthyaGrid

A voice-first control room for district medicine supply. It flags the clinic shelf that runs out next, proposes the fix inside the district, and answers the officer aloud in Hindi, Hinglish or English on Gnani's speech and language models.

Built by	Institution	Role
Harsh Kawatra	Delhi Technological University	Product, code

Live app ↗ · Source code ↗ · Walkthrough video ↗

The medicine an empty shelf needs is usually already in the district, at a warehouse or a clinic a short drive away, but nobody adds the stock up each morning. SwasthyaGrid does that sum and proposes the move. A named officer approves it, and the language model never computes a number.

1.36 crore
stock-out days avoided a year, base estimate

72.2% → 86.1%
key-medicine availability, before and after

1 Problem: up to a third of key medicines can be missing from a rural shelf, and the district hears about it late

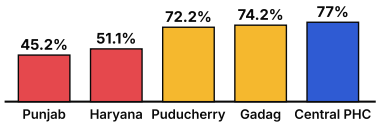
45 to 80%
key medicines in stock, by state study

38,241
PHCs and CHCs hold the rural shelf

43.4%
of health spending paid out of pocket

79.9%
CHC specialist posts unfilled

KEY-MEDICINE AVAILABILITY, BY STUDY

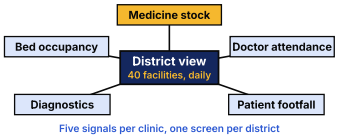


No study finds more than 80%. India records 58,000 snakebite deaths a year.

Root cause: stock is recorded at every clinic, never added up district-wide

- **Seen late:** HMIS reports monthly, after the shelf empties
- **Never moved:** surplus at a nearby clinic sits unused
- **No forecast:** burn rates and bed pressure go uncounted
- **Wrong interface:** typed English dashboards for Hindi speakers

2 Product: one loop from flag to delivery, computed by plain arithmetic, with a named officer approving every move



Five signals per clinic, one screen per district

1 See
Command Centre scores each facility on five signals, with a risk map and an AI briefing

2 Flag
Days of cover for every medicine at every facility, ranked by risk

3 Propose
A warehouse refill, or a nearby transfer, with reasons and a confidence score

4 Approve
The officer approves, edits or rejects. Nothing ships without that click

5 Move
Truck and driver assigned, cold chain for vaccines, tracked on real roads

6 Confirm
The clinic confirms receipt on a phone and risk drops on every screen

Rules an auditor can recompute: days of cover = units ÷ average daily use · flag under 5 days · critical under 3, or under 5 for emergency medicines · refill to 21 days · transfer within 40 km · bed redirect above 90% · confidence 40 to 98

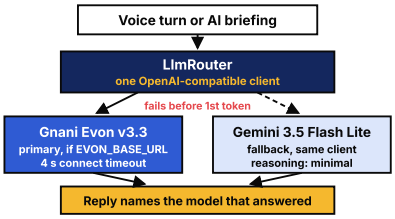
3 Voice: every spoken turn runs on Gnani, from the officer's Hinglish question to the answer they hear

Officer speaks (16 kHz PCM, browser) → **Prisma v2.5 STT** (Gnani, WebSocket) → **LLM router + tools** (Evon v3.3, then Gemini) → **Sentence split** (as tokens stream) → **Timbre v2.5 TTS** (Gnani, WebSocket) → **Speaker** (24 kHz PCM)

Stage	Gnani model	How SwasthyaGrid uses it
Speech to text	Prisma v2.5	Mic audio is repacked into the exact 1024-byte frames Gnani requires. One final transcript per utterance shows on screen and confirms a barge-in at two words. English and 10 Indian languages; **prisma-v3.0** adds language detection.
Language model	Evon v3.3 first	Up to 3 rounds over 13 read-only tools on the dashboard's own services, so spoken and typed answers agree. A cached filler, "One moment, checking the data.", covers tool time.
Text to speech	Timbre v2.5	Streams 24 kHz audio a sentence at a time, since Gnani allows one active synthesis per key. Kaveri speaks English, Nalini Hindi. A dropped socket finishes the turn over REST.

Probed on Gnani's Vachana API, 8 Nov 2026: four voices returned audio in 3.0 to 3.9 s; first streamed audio 2.4 s after a cold socket; Hindi and English transcripts closely matched; a second simultaneous synthesis got HTTP 429, so sentences are queued.

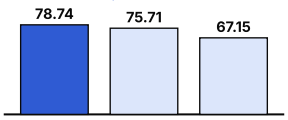
4 Language model: Evon v3.3 is the primary model by one setting, and Gemini answers whenever Evon is unset or down



Rule	How it works in the code
Swap	**EVON_BASE_URL** (vLLM, SGLang or HF endpoint) makes Evon primary; **LLM_PRIMARY** flips the order
Failover	Before the first token, never mid-answer; a failed provider is parked 60 s
Clean output	Strips reasoning traces; keeps Gemini's tool-call signatures
Disclosure	Voice metrics and the briefing name the model that answered
Grounding	Engines compute every number; the model only phrases it

The hosted app answers on Gemini today, because Evon needs an 80 GB-class GPU.

MILU ACCURACY, 11-LANGUAGE MEAN



Hindi: 81.99 vs 76.95 vs 67.08

Evon: 30B mixture of experts, about 3.5B active, 128K context, Apache-2.0.

5 Impact: an estimated 1.36 crore stock-out days avoided

1.36 crore stock-out days avoided a year at 38,241 PHCs and CHCs; base case, half of eVIN's 80% cut. A model estimate

2.72 crore
stock-out days a year now

72.2% → 86.1%
key-medicine availability

17,412
per district a year

Bear, 30% cut — 0.81 crore
Base, 50% cut — 1.36 crore
Bull, 80% cut — 2.17 crore

6 Built: a working, open-source prototype on simulated data

Item	Where it stands
Stack	Next.js 16 on Vercel; FastAPI on Cloud Run, Mumbai
Scope	40 facilities, 5 Rajasthan districts; simulated stock, labelled so
Roads	365 real OpenStreetMap routes, none synthetic
One run	PHC Kota-4 anti-rabies vaccine: 1.2 to 21.0 days of cover after one approved refill
Tests	386 backend and 271 frontend tests in CI on every push
Next	Evon on a GPU, its tool calling run through the voice evaluation; then a shadow-mode district

Sources: availability, PMC8654139, PMC10292169 and Kotwani 2015 · out-of-pocket share, NHA 2022-23 · snakebite, eLife 2020 · facilities and specialist posts, MoHFW HDI 2022-23 · eVIN, UNDP India · Evon benchmarks, Hugging Face model card gnani/gnani-evon-v3.3-30B-A3B · Gnani API behaviour, SwasthyaGrid probe of 8 Nov 2026. **Impact model:** 38,241 facilities × 7 tracer medicines × 365 days × 27.8% unavailability × a 30%, 50% or 80% cut; about 780 districts. Estimates, not measured results.